

Mohammed Amaan Ahmed Khan

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EDUCATION

Arizona State University

Master of Science in Data Science and Analytics | GPA: 3.5/4.0

Tempe, AZ

Aug. 2024 – May 2026

SRM Institute of Science and Technology

Bachelor of Technology in Computer Science, Specialization in AI and ML | GPA: 3.7/4.0

July 2018 – May 2022

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript/TypeScript, SQL

Backend & Systems: Distributed Systems, Parallel Systems, Large-scale System Design, Microservices, REST APIs, GraphQL, Networking, Unix/Linux, Information Retrieval

Cloud & Infrastructure: AWS (EC2, S3, Lambda), GCP (BigQuery, Firebase), Docker, Kubernetes, Terraform, CI/CD

Data & Databases: PostgreSQL, NoSQL, ETL Pipelines, Data Modeling, Vector Search, Embeddings

Machine Learning & AI: TensorFlow, scikit-learn, XGBoost, Generative AI, Agentic AI, LLMs, RAG, NLP, Computer Vision

Frameworks & Tools: Node.js, Express.js, React.js, pandas, NumPy, Git, Jenkins

EXPERIENCE

Software Engineer II

July 2022 – August 2024

Verizon

- Engineered, deployed, and maintained a distributed **Python** and **Java** backend for **6,300+** users, improving platform reliability by **30%** with fault isolation across **Unix/Linux** environments.
- Designed, built, and tested full-stack **REST APIs** with end-to-end **CI/CD** coverage across **3+** microservices, automating release regressions down by **20%**.
- Operated **3,000+** service migration end-to-end with phased rollout and validation gates, reducing incidents by **25%** and MTTR by **20%**.
- Developed and shipped an **NLP**-based chatbot powered by **LLM** inference into production, trimming support response time by **60%**.
- Built and maintained scalable **JavaScript** and **Python** web applications on **Unix/Linux**, improving order processing efficiency by **40%**.

AI Full Stack Engineer

June 2021 – September 2021

Banao Technologies

- Debugged and resolved **15+** critical bottlenecks across **distributed systems** on rotational on-call, owning fixes end-to-end and improving system performance by **30%**.
- Designed and deployed an **ML**-powered **ranking** and **prediction** pipeline end-to-end, replacing legacy rule-based systems and cutting manual intervention by **70%**.
- Identified, triaged, and rectified **50+** production bugs across **3+** services, stabilizing system reliability within project deadlines.

Software Developer

February 2021 – September 2021

Vidya (Formerly Across The Globe)

- Developed and maintained **JavaScript** dashboards with parallel **Python** data pipelines, improving analytics access for **3+** teams.
- Programmed and optimized **information retrieval** pipelines using **PostgreSQL**, reducing data retrieval latency by **25%** for public health analytics.

PROJECTS

KG-Grounded RAG for Consistent LLM Reasoning | Python, Neo4j, LangChain, OpenAI

Jan 2026 - March 2026

- Boosted grounded-answer accuracy by **11%** on **1,000+** evaluations by pioneering a **Generative AI** system using **RAG**, **NLP**, and **knowledge graph** retrieval constraints.
- Suppressed contradictory **LLM** responses by **22%**, formulating **ML** ranking pipelines over vanilla **RAG** with verifier correction.

Distributed API Test Automation for Containerized Services | Python, C#, Docker

August 2025 - December 2025

- Parallelized **REST** validation across **Docker** containers by constructing a multi-threaded **distributed systems** test runner, cutting execution time by **20%**.
- Eradicated flaky runs by instituting deterministic data seeding as a hard **CI/CD** deployment validation gate.

Road Scene Analysis using DeepLabv3+ | Python, TensorFlow, Cityscapes

December 2024 – May 2025

- Attained **82%** IoU on **Cityscapes** by training a **Computer Vision** segmentation model using **TensorFlow** and **ML** infrastructure.
- Compressed inference latency by **35%** via quantization and model compression across **ML** pipelines for edge deployment.